

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eight Semester of B. Tech. Examination (EE)

May 2014

EE 406 Electrical Machine Design-II

Date: 05.05.2014 Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 A 1250 kVA, 3.3 kV, 50 Hz, 300 rpm, 3 phase star connected alternator has 180 slots and 5 conductors per slot having single layer winding with full pitch coils and one circuit per phase. Determine the specific electric and specific magnetic loadings if the air gap diameter of alternator is 2.2 m and axial length is 0.5 m. Assume phase spread of 60° . [07]

Q - 2 (a) Define the specific magnetic loading and explain the factors influencing the selection of specific magnetic loading for designing a three phase alternator [07]

OR

Q - 2 (a) Define the specific electric loading and explain the factors influencing the selection of specific electric loading for designing a three phase alternator. [07]

Q - 2 (b) Write the equation to calculate the length of air gap of an alternator. Discuss the effect of air gap length on the performance of a synchronous machine. [07]

Q - 3 (a) Design the field coil of a 3 phase, 16 poles, 50 Hz salient pole alternator, based on the following design information. Stator diameter at gap surface = 1.0 m, Gross length of stator core = 0.3 m, Section of pole body = $0.15 \text{ m} \times 0.3 \text{ m}$, Height of the pole = 0.15 m, Ampere turns per pole = 6500, Exciter voltage = 110 V. Assume suitable data where ever necessary. [07]

OR

Q - 3 (a) Describe in steps the complete procedure for the design of field winding of an alternator. [07]

Q - 3 (b) A 1250 kVA, 3 phase, 6600 V, salient pole alternator has the following data: Air gap diameter 1.6 m, length of core 0.45 m, number of poles 20, armature ampere conductors per metre 28000, ratio of pole arc to pole pitch 0.68, stator slot pitch 28 mm, current density in damper bars 3 A/mm². Design a suitable damper winding for the machine. [07]

OR

Q - 3 (b) Explain the procedure for design of the damper winding of a 3 phase salient pole generator. [07]

SECTION – II

- Q - 4 Calculate the following design information for a 30 kW, 440 V, 3 phase, 6 pole, 50 Hz, delta connected, squirrel cage induction motor, (i) Main dimensions of stator frame (ii) Number of turns per phase in stator winding (iii) Number of stator slots (iv) Number of conductors per slot. Assume $B_{av} = 0.48$ Tesla, $a_c = 26000$ ac/m, efficiency 88 % and power factor 0.86 lagging, $K_w = 0.955$, design for best power factor operation. [07]
- Q - 5 (a) Explain the dispersion coefficient and its effect on overload capacity of an induction machine. [07]
- Q - 5 (b) Draw the current distribution in squirrel cage rotors and derive equation for end ring current and area of end ring.

OR

- Q - 5 (b) A 3 phase 2 pole 50 Hz squirrel cage induction motor has a rotor diameter 0.20 m and core length 0.12 m. The peak density in the air gap is 0.55 Tesla. The rotor has 33 bars, each of resistance $125 \mu\Omega$ and a leakage inductance $2 \mu H$. The slip is 6 %. Calculate (i) the peak value of current in each bar (ii) rotor I^2R loss (iii) rotor output and (iv) torque exerted. Neglect the resistance of end rings. [07]
- Q - 6 (a) Derive the equation of capacitor for maximum torque in single phase capacitor start induction motor. [07]

OR

- Q - 6 (a) Derive the relation between D and L for the best power factor in case of induction motor. [07]
- Q - 6 (b) Draw the overall flow chart for overall design of induction machine. [07]

OR

- Q - 6 (b) A 3 phase, 3000V, 260kW, 50Hz, 10 pole squirrel cage induction motor gave the following results during its preliminary design. Internal diameter of stator = 75cm, Gross length of stator = 35cm, Number of stator slots = 125, Number of conductors per slot = 10. Based on the above data calculate the following for the squirrel cage rotor. (a) Total losses in the rotor bars (b) Losses in the end rings (c) Equivalent resistance of the rotor in terms of stator. [07]
